

Separability and the limits of multivariate forecast reconciliation

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Abstract

When several variables are observed over the same hierarchy, such as worker admissions and dismissals recorded by region, it seems natural to reconcile their forecasts jointly so that cross-variable dependence can inform the adjustment. We show when this helps. Reading minimum trace reconciliation as a regression of the base forecast errors on the observed incoherences, joint reconciliation coincides with reconciling each variable separately if and only if each variable's separately reconciled errors are uncorrelated with the other variables' incoherences. Separable error covariances, and dependence confined to the coherent part of the errors, are special cases, so in a common simulation design joint reconciliation has no population advantage at all. We propose two effect sizes and a likelihood ratio test for real data. In simulations the gains are modest even when the error covariance is known, and with an estimated covariance joint reconciliation can lose accuracy at small sample sizes because it must estimate cross-variable covariances. For monthly Brazilian employment data over 2007–2019, the test detects a small departure, and with ARIMA base forecasts joint reconciliation does not systematically improve the forecasts of either variable. It does improve forecasts of net employment change, reducing the national mean squared error by 18%, consistent with the fact that with the true covariance minimum trace reconciliation is optimal for every linear combination of the series, not just on average. For probabilistic forecasts of net change, modelling the dependence between the variables' base forecasts matters most.

Keywords: Hierarchical time series, forecast reconciliation, minimum trace, separable covariance, employment forecasting

1. Introduction

Collections of time series can often be aggregated at several levels by geography, product, or other attributes. Forecasts are usually needed for every series at every level, and they should be coherent: each aggregate forecast should equal the sum of the forecasts below it. Forecast reconciliation achieves this by forecasting every series independently and then adjusting the forecasts to satisfy the aggregation constraints (Hyndman et al., 2011; Athanasopoulos et al., 2024). The minimum trace (MinT) method of Wickramasuriya et al. (2019) chooses the adjustment using the covariance matrix of the base forecast errors, and it is now the standard approach.

Reconciliation is almost always applied to one variable at a time, yet many hierarchies carry several variables. Brazil records monthly worker admissions and dismissals for each of its 27 federative units, which aggregate to five regions and to the country. The two variables are driven by common labour market conditions, and their forecast errors are correlated. Because MinT works by exploiting the covariance of the base forecast errors, it seems natural to reconcile both variables together, using a covariance matrix that spans them, and to expect this to beat reconciling each separately. This paper asks when that expectation is correct.

Joint reconciliation of several variables is not the same problem as reconciling a grouped hierarchy. In a grouped hierarchy, such as tourism demand disaggregated by both region and travel purpose, the categories of each grouping variable are linked by aggregation constraints: total tourism is the sum over purposes, and that total is itself forecast. Here nothing sums admissions to dismissals. The stacked summing matrix is block diagonal by variable, and the only link between the variables is the error covariance.

We give an exact answer. MinT can be read as a regression of the base forecast errors on the observed incoherences of the base forecasts. Joint reconciliation of several variables coincides with separate reconciliation if and only if, once each variable has been reconciled on its own, its remaining errors are uncorrelated with the incoherences of every other variable. Two known results are special cases. MinT is unaffected by any component of the error covariance that lies in the coherent subspace (Wickramasuriya, 2021), so cross-variable dependence of that form is invisible. And when the error covariance is separable, a Kronecker product of a cross-variable matrix and a cross-series matrix, joint and separate reconciliation coincide whatever the cross-variable correlation, a special case of a result of Girolimetto and Di Fonzo (2025) for cross-temporal reconciliation and the reconciliation analogue of Zellner’s observation that seemingly unrelated regressions with identical regressors reduce to equation-by-equation least squares (Zellner, 1962).

The characterisation has an immediate consequence for how joint reconciliation has been evaluated. A natural simulation design generates correlated innovations for the bottom-level series of every variable and gives every node the same dynamics. We show that the population-optimal forecasts in such designs are coherent, so reconciliation of any kind has nothing to act on, and differences between joint and separate reconciliation arise only from estimation error in the base models. Varying the cross-variable correlation in such a design says nothing about the value of joint reconciliation.

For real data we need to know whether the conditions of the characterisation fail, and by how much it matters. Measures of distance from a Kronecker product answer the wrong question, because they respond to components of the covariance that reconciliation ignores. We propose two effect sizes: κ , the weight that the joint reconciliation places on other variables’ incoherences, and γ , the reduction in mean squared error that it would deliver if the estimated covariance were correct. We also propose a likelihood ratio test of the characterising condition, which uses the regression form of MinT and needs no simulation.

Two experiments show how little joint reconciliation offers. When the error covariance is known, even substantial departures from separability reduce the mean squared error by at most 15.7% in our designs, and usually by much less. When it is estimated, joint reconciliation pays for the extra cross-variable blocks: with separable errors it is worse than separate reconciliation at small sample sizes, and it recovers small population gains only with long series. With fitted ARIMA and VAR base models, joint and separate reconciliation are indistinguishable for series of typical monthly length. In the Brazilian data, the test detects a small departure from the conditions of the characterisation, but with ARIMA base forecasts joint reconciliation does not systematically improve the forecasts of either admissions or dismissals.

Joint modelling does matter for quantities that combine the variables. Net employment change, the difference between admissions and dismissals, is a key quantity for users of these data. MinT with the true covariance is optimal in the positive semidefinite order, so when the conditions of the characterisation fail, joint reconciliation can help a combination of variables even when it does not help any variable on its own. In the application it reduces the mean squared error of the national net change by 18%. For probabilistic forecasts the larger effect comes earlier: simulating the base forecasts jointly, so that the dependence between the variables is kept, reduces the CRPS of the national net change by 15%, and reconciling the paths jointly raises this to 21%.

Section 2 sets out notation and the reconciliation problem for several variables. Section 3 gives the characterisation and its consequences, with proofs in the Appendix. Section 4 develops the effect sizes

and the test. Section 5 reports the two experiments, Section 6 the application to Brazilian employment, and Section 7 discusses the implications.

2. Reconciling several variables

2.1. Notation

We follow the notation of Athanasopoulos et al. (2024), adapted to several variables. Suppose m variables are each observed over the same hierarchy with n series, of which n_b are at the bottom level and $n_a = n - n_b$ are aggregates. For variable j , let $\mathbf{y}_{j,t}$ be the n -vector of all its series at time t , and $\mathbf{b}_{j,t}$ the n_b -vector of its bottom-level series. Then

$$\mathbf{y}_{j,t} = \mathbf{S}\mathbf{b}_{j,t}, \quad \mathbf{S} = \begin{bmatrix} \mathbf{A} \\ \mathbf{I}_{n_b} \end{bmatrix}, \quad (1)$$

where the $(n_a \times n_b)$ matrix $\mathbf{A}$ gives the aggregation constraints. Equivalently, $\mathbf{C}\mathbf{y}_{j,t} = \mathbf{0}$ with $\mathbf{C} = [\mathbf{I}_{n_a} \quad -\mathbf{A}]$, so that $\mathbf{C}\mathbf{S} = \mathbf{0}$. Figure 1 shows a three-level example with $n = 8$ and $n_b = 5$.

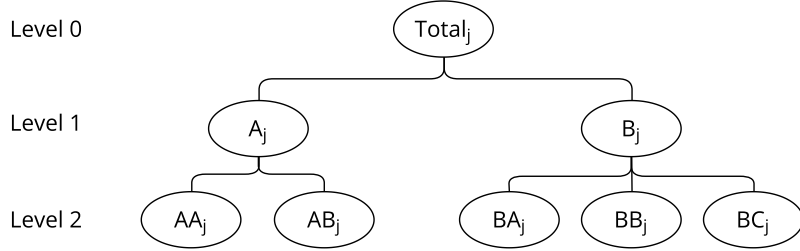


Figure 1: A hierarchy with three levels, observed for each of m variables.

Collect the variables in $\mathbf{Y}_t = [\mathbf{y}_{1,t}, \dots, \mathbf{y}_{m,t}]$ and $\mathbf{B}_t = [\mathbf{b}_{1,t}, \dots, \mathbf{b}_{m,t}]$. Stacking the columns gives

$$\text{vec}(\mathbf{Y}_t) = \mathbf{S}_* \text{vec}(\mathbf{B}_t), \quad \mathbf{S}_* = \mathbf{I}_m \otimes \mathbf{S}, \quad (2)$$

with constraint matrix $\mathbf{C}_* = \mathbf{I}_m \otimes \mathbf{C}$. The stacked summing matrix is block diagonal: no constraint links one variable to another. This distinguishes the setting from a grouped hierarchy, where the categories of a grouping variable (such as travel purpose in the Australian tourism data of Wickramasuriya et al., 2019) are linked by aggregation constraints because their total is itself observed and forecast.

2.2. Forecast reconciliation

Let $\hat{\mathbf{y}}_{T+h|T}$ be the stacked h -step base forecasts of all mn series, produced without regard to the constraints, and let $\mathbf{W}_h$ be the covariance matrix of their errors. Linear reconciliation maps the base forecasts to coherent forecasts $\tilde{\mathbf{y}}_{T+h|T} = \mathbf{S}_* \mathbf{P} \hat{\mathbf{y}}_{T+h|T}$ for some matrix $\mathbf{P}$ with $\mathbf{P}\mathbf{S}_* = \mathbf{I}$. The minimum trace (MinT) choice (Wickramasuriya et al., 2019) minimises the trace of the reconciled error covariance and gives

$$\tilde{\mathbf{y}}_{T+h|T} = \mathbf{S}_* (\mathbf{S}_*' \mathbf{W}_h^{-1} \mathbf{S}_*)^{-1} \mathbf{S}_*' \mathbf{W}_h^{-1} \hat{\mathbf{y}}_{T+h|T} = [\mathbf{I} - \mathbf{W}_h \mathbf{C}_*' (\mathbf{C}_* \mathbf{W}_h \mathbf{C}_*)^{-1} \mathbf{C}_*] \hat{\mathbf{y}}_{T+h|T}, \quad (3)$$

the second form being the zero-constrained representation of Di Fonzo and Girolimetto (2023). Applied to the stacked system this is ordinary MinT with summing matrix $\mathbf{S}_*$; we call it *joint* reconciliation. *Separate* reconciliation applies MinT to each variable on its own, using only the diagonal block of $\mathbf{W}_h$ for that variable. Joint reconciliation can use the cross-variable blocks of $\mathbf{W}_h$, and it is natural to expect it to be more accurate when the variables' forecast errors are correlated. Section 3 shows when that expectation holds.

In practice $\mathbf{W}_h$ is unknown. Following Wickramasuriya et al. (2019), we set $\mathbf{W}_h = k_h \widehat{\mathbf{W}}_1$, where $\widehat{\mathbf{W}}_1$ is the shrinkage estimator

$$\widehat{\mathbf{W}}_1 = \lambda \widehat{\mathbf{W}}_{1,D} + (1 - \lambda) \widehat{\mathbf{W}}_{1,S}, \quad (4)$$

$\widehat{\mathbf{W}}_{1,S}$ is the sample covariance matrix of the one-step in-sample residuals, $\widehat{\mathbf{W}}_{1,D}$ is its diagonal, and the intensity λ is estimated as in [Schäfer and Strimmer \(2005\)](#). The scale k_h cancels from (3). For separate reconciliation the estimator is applied to each variable's residuals on its own, so each variable has its own shrinkage intensity.

3. When does joint reconciliation matter?

3.1. Joint and separate reconciliation

Stack the m variables as in (2), so that the summing and constraint matrices of the stacked system are $\mathbf{S}_* = \mathbf{I}_m \otimes \mathbf{S}$ and $\mathbf{C}_* = \mathbf{I}_m \otimes \mathbf{C}$. Let $\mathbf{W}$ be the $(mn \times mn)$ covariance matrix of the stacked base forecast errors $\mathbf{e} = (\mathbf{e}'_1, \dots, \mathbf{e}'_m)'$, where $\mathbf{e}_j$ holds the n errors of variable j , and write $\mathbf{W}_{jk} = \text{Cov}(\mathbf{e}_j, \mathbf{e}_k)$ for its $(n \times n)$ blocks. We drop the horizon subscript h throughout this section.

Joint MinT reconciliation in zero-constrained form ([Di Fonzo and Girolimetto, 2023](#)) is $\tilde{\mathbf{y}} = \mathbf{M}\hat{\mathbf{y}}$ with

$$\mathbf{M} = \mathbf{I} - \mathbf{G}\mathbf{C}_*, \quad \mathbf{G} = \mathbf{W}\mathbf{C}'_*(\mathbf{C}_*\mathbf{W}\mathbf{C}'_*)^{-1}. \quad (5)$$

Separate reconciliation applies univariate MinT to each variable using its own error covariance $\mathbf{W}_{jj}$. Its map is block diagonal, $\mathbf{M}^{\text{sep}} = \text{diag}(\mathbf{M}_1, \dots, \mathbf{M}_m)$, with

$$\mathbf{M}_j = \mathbf{I} - \mathbf{G}_j\mathbf{C}, \quad \mathbf{G}_j = \mathbf{W}_{jj}\mathbf{C}'(\mathbf{C}\mathbf{W}_{jj}\mathbf{C}')^{-1}. \quad (6)$$

Both maps have a regression interpretation. The incoherence of the base forecasts, $\mathbf{C}_*\hat{\mathbf{y}} = -\mathbf{C}_*\mathbf{e}$, is observable, and $\mathbf{G}$ is the matrix of population regression coefficients of the errors $\mathbf{e}$ on the incoherences $\mathbf{C}_*\mathbf{e}$, since $\text{Cov}(\mathbf{e}, \mathbf{C}_*\mathbf{e}) = \mathbf{W}\mathbf{C}'_*$. MinT adds the fitted value of this regression to the base forecasts: $\tilde{\mathbf{y}} = \hat{\mathbf{y}} + \mathbf{G}\mathbf{C}_*\mathbf{e}$, an estimate of the error made from the incoherences, which are observed. Separate reconciliation runs the same regression for each variable, using only its own incoherences $\mathbf{C}\mathbf{e}_j$. The off-diagonal blocks of $\mathbf{G}$ are the coefficients on the other variables' incoherences, and joint reconciliation can only improve on separate reconciliation through them.

3.2. A characterisation

Proposition 1. *Let $\mathbf{W}$ be positive definite. The following statements are equivalent.*

- (a) $\mathbf{M} = \mathbf{M}^{\text{sep}}$: joint and separate reconciliation give the same forecasts for every base forecast.
- (b) $\mathbf{M}$ is block diagonal by variable.
- (c) $\mathbf{G}$ is block diagonal by variable.
- (d) $\mathbf{M}_j\mathbf{W}_{jk}\mathbf{C}' = \mathbf{0}$ for all $j \neq k$; that is, $\text{Cov}(\mathbf{M}_j\mathbf{e}_j, \mathbf{C}\mathbf{e}_k) = \mathbf{0}$.
- (e) The column space of $\mathbf{W}_{jk}\mathbf{C}'$ is contained in that of $\mathbf{W}_{jj}\mathbf{C}'$ for all $j \neq k$.

Condition (d) has a direct reading. Joint reconciliation changes nothing exactly when, after each variable has been reconciled on its own, its remaining errors are uncorrelated with the incoherences of every other variable. Condition (b) shows that there is no intermediate case. If the reconciled forecasts of one variable depend on the base forecasts of another at all, they differ from the separately reconciled forecasts.

Two known results are special cases of Proposition 1.

Corollary 1 (Invariance). *For any $\mathbf{K}$ such that $\mathbf{W} + \mathbf{S}_*\mathbf{K}\mathbf{S}'_*$ is positive definite, replacing $\mathbf{W}$ by $\mathbf{W} + \mathbf{S}_*\mathbf{K}\mathbf{S}'_*$ leaves $\mathbf{M}$, $\mathbf{M}^{\text{sep}}$ and $\mathbf{G}$ unchanged.*

This is the argument used by [Wickramasuriya \(2021, Proposition 1\)](#) to show that the GLS reconciliation of [Hyndman et al. \(2011\)](#) and MinT coincide. It is an instance of the classical result that generalised least squares is unaffected by adding $\mathbf{X}\mathbf{A}\mathbf{X}'$ to the error covariance ([Rao, 1967](#); [Zyskind, 1967](#)). The implication here is that only the part of the error covariance that is visible through the incoherences matters. Any cross-variable dependence of the form $\mathbf{S}\mathbf{K}_{jk}\mathbf{S}'$ is invisible to both joint and separate reconciliation.

Corollary 2 (Separability). *If $\mathbf{W} = \mathbf{V} \otimes \mathbf{\Sigma}_0$ for positive definite matrices $\mathbf{V}$ ($m \times m$) and $\mathbf{\Sigma}_0$ ($n \times n$), then $\mathbf{M} = \mathbf{M}^{\text{sep}} = \mathbf{I}_m \otimes \mathbf{M}_0$, where $\mathbf{M}_0 = \mathbf{I} - \mathbf{\Sigma}_0 \mathbf{C}' (\mathbf{C} \mathbf{\Sigma}_0 \mathbf{C}')^{-1} \mathbf{C}$, whatever the sign or strength of the cross-variable dependence in $\mathbf{V}$.*

Corollary 2 is the special case of a result of [Girolimetto and Di Fonzo \(2025\)](#) for cross-temporal reconciliation in which the second dimension carries no aggregation constraints. It is the reconciliation analogue of the result that seemingly unrelated regressions with identical regressors reduce to equation-by-equation least squares ([Zellner, 1962](#)): every variable shares the same summing matrix, so cross-variable correlation alone cannot be exploited. Corollaries 1 and 2 combine: every $\mathbf{W} = \mathbf{V} \otimes \mathbf{\Sigma}_0 + \mathbf{S}_* \mathbf{K} \mathbf{S}_*'$ satisfies the conditions of Proposition 1, however far it is from a Kronecker product. The same invariance gives the known condition $\mathbf{W} = \mathbf{S}_* \mathbf{K} \mathbf{S}_*' + \tau \mathbf{I}$ under which MinT reduces to OLS reconciliation ([Hyndman et al., 2011](#); [Wickramasuriya, 2021](#)).

3.3. Forecast errors from node-invariant dynamics

The structure of $\mathbf{W}$ is determined by how the base forecasts are produced, and a common simulation design produces a covariance that reconciliation cannot use at all.

Proposition 2. *Let $\mathbf{x}_{i,t}$ be the m -vector of the bottom-level series at node i , with $\mathbf{x}_{i,t} = \boldsymbol{\mu}_{i,t} + \boldsymbol{\eta}_{i,t}$, where $\boldsymbol{\mu}_{i,t}$ is deterministic and $\boldsymbol{\eta}_{i,t} = \boldsymbol{\Phi} \boldsymbol{\eta}_{i,t-1} + \boldsymbol{\varepsilon}_{i,t}$ is a stationary VAR(1) with the same $\boldsymbol{\Phi}$ at every node. Suppose the innovations are serially uncorrelated with $\text{Cov}(\boldsymbol{\varepsilon}_{i,t}, \boldsymbol{\varepsilon}_{i',t}) = \sigma_{ii'} \mathbf{R}$ for an $(m \times m)$ matrix $\mathbf{R}$ and an $(n_b \times n_b)$ matrix $\boldsymbol{\Sigma} = [\sigma_{ii'}]$, as in the simulation design of Section 5. Let the base forecast of each series be the best linear predictor of its stochastic component from its own past, plus its deterministic component. Then the base forecasts are coherent and the base forecast errors lie in the column space of $\mathbf{S}_*$. Their covariance is singular, so the MinT formula (5) is not defined, but every linear reconciliation $\mathbf{S}_* \mathbf{P}$ with $\mathbf{P} \mathbf{S}_* = \mathbf{I}$, joint or separate, leaves coherent forecasts unchanged.*

Proposition 2 shows that in such designs the population error covariance is entirely of the form $\mathbf{S}_* \mathbf{K} \mathbf{S}_*'$. The differences between reconciliation methods seen in simulations then come only from the base models: their estimation error, and the different model specifications selected for different series. What matters for joint reconciliation is how this incoherent component of the base forecast errors is correlated across variables. The component is small when the base models are good, and its cross-variable structure is hard to estimate. Sections 4 and 5 quantify both points.

3.4. Remarks

Horizon. Under the usual assumption $\mathbf{W}_h = k_h \mathbf{W}_1$, the maps $\mathbf{M}$ and $\mathbf{M}^{\text{sep}}$ do not depend on h , and neither do the conditions of Proposition 1.

Shrinkage. The diagonal target of the shrinkage estimator has zero cross-variable blocks and so satisfies condition (d). Full shrinkage therefore makes joint reconciliation identical to reconciling each variable separately with its own diagonal target, and partial shrinkage scales the estimated cross-variable blocks of $\mathbf{W}$ towards zero.

Linear combinations. MinT minimises not only the trace of the reconciled error covariance but the whole matrix in the positive semidefinite order, since it is the generalised least squares projection ([Wickramasuriya et al., 2019](#)). Separate reconciliation is another unbiased linear projection, so for every linear combination $\mathbf{a}' \mathbf{y}$ of the stacked series, such as the difference between two variables at one node, joint reconciliation with the true $\mathbf{W}$ has no larger error variance. When the conditions of Proposition 1 fail, the gain need not be spread evenly: it can be negligible for each variable's own series yet substantial for combinations that load on several variables.

More than two variables. Proposition 1 holds for any m , and condition (d) can be checked pair by pair. Joint reconciliation of m variables can be split into separate reconciliations of groups of variables whenever the off-diagonal blocks of $\mathbf{G}$ between groups vanish.

4. Measuring whether joint reconciliation can help

Proposition 1 gives an exact condition, but in practice $\mathbf{W}$ is estimated and the condition never holds exactly in the estimate. We need a measure of how far an estimated covariance is from satisfying it, and a way to judge whether the observed departure is larger than estimation noise alone would produce.

4.1. Separability is the wrong target

The natural first choice is a measure of distance from a Kronecker product, such as the relative error of the nearest Kronecker approximation (Van Loan and Pitsianis, 1993). Rearranging $\mathbf{W}$ so that each block $\mathbf{W}_{jk}$ becomes a row, $\mathbf{W}$ is a Kronecker product exactly when the rearranged matrix has rank one, and the relative error of the nearest Kronecker product is $\{1 - \sigma_1^2 / \sum_i \sigma_i^2\}^{1/2}$, where σ_i are its singular values. Likelihood ratio tests of separability are also available (Lu and Zimmerman, 2005; Mitchell et al., 2006).

These measures answer the wrong question. By Corollary 1, reconciliation ignores any component of $\mathbf{W}$ of the form $\mathbf{S}_* \mathbf{K} \mathbf{S}_*'$, and such components can make $\mathbf{W}$ arbitrarily far from a Kronecker product without changing the reconciled forecasts at all. Conversely, departures that do change the forecasts may leave the Kronecker error small. Figure 2 plots the gain from joint reconciliation against the nearest-Kronecker error for the covariance families of Section 5. The Kronecker error rises with no gain at all when the departure is invisible, and it can fall while the gain rises.

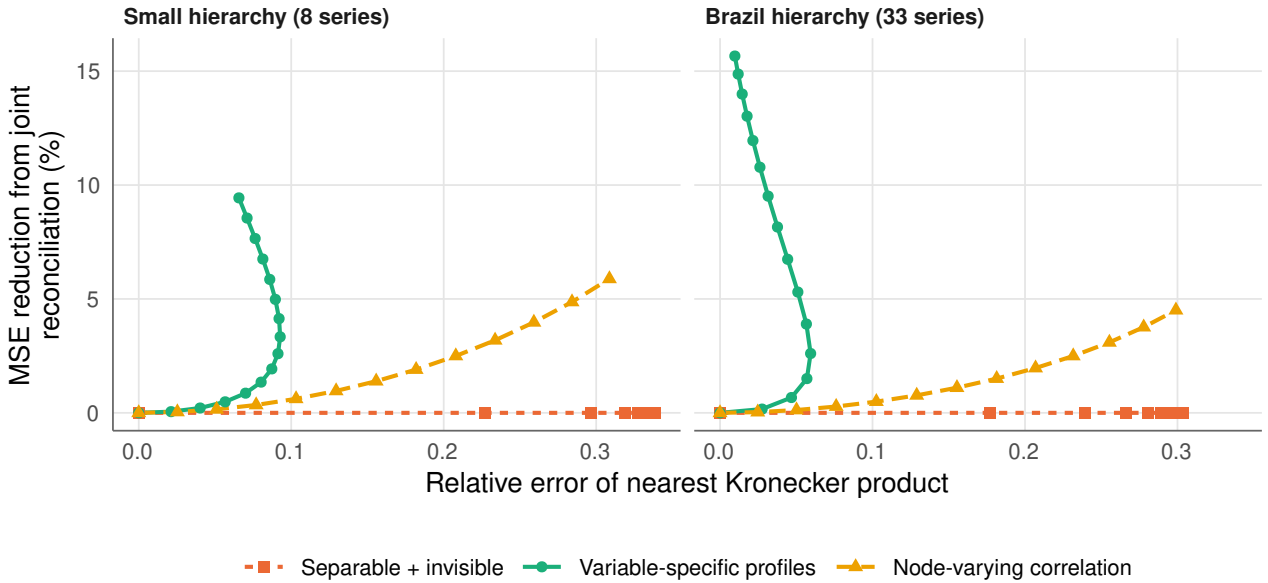


Figure 2: Population MSE reduction from joint reconciliation against the relative error of the nearest Kronecker approximation to $\mathbf{W}$, for the covariance families of Section 5.

4.2. Two measures

We instead measure the conditions of Proposition 1 directly. The first measure describes the structure of the joint map. Since $\mathbf{G}$ contains the regression coefficients of the base forecast errors on the incoherences, and its off-diagonal blocks are the coefficients on other variables' incoherences, we define

$$\kappa(\mathbf{W}) = \left(\frac{\sum_{j \neq k} \|\mathbf{G}_{jk}\|_F^2}{\|\mathbf{G}\|_F^2} \right)^{1/2}, \quad (7)$$

where $\mathbf{G}_{jk}$ is the $(n \times n_a)$ block of $\mathbf{G}$ relating the errors of variable j to the incoherences of variable k . Rescaling variable k by c_k multiplies $\mathbf{G}_{jk}$ by c_j/c_k , so before computing κ we standardise each variable by the square root of the mean diagonal of $\mathbf{W}_{jj}$. This makes κ free of units. By Proposition 1,

$\kappa = 0$ exactly when joint and separate reconciliation coincide. Adding a component $\mathbf{S}_* \mathbf{K} \mathbf{S}'_*$ to $\mathbf{W}$ leaves $\mathbf{G}$ unchanged (Corollary 1), so it cannot move κ away from zero, although it can change the standardisation and hence the value of a non-zero κ .

The second measure describes the value of the joint map. If $\mathbf{W}$ were the true error covariance, the proportional reduction in mean squared error from reconciling jointly rather than separately would be

$$\gamma(\mathbf{W}) = 1 - \frac{\text{tr}(\mathbf{M} \mathbf{W} \mathbf{M}')}{\text{tr}(\mathbf{M}^{\text{sep}} \mathbf{W} \mathbf{M}^{\text{sep}'})}. \quad (8)$$

MinT minimises the trace over all maps of this form, so $\gamma \geq 0$, again with equality exactly when the conditions of Proposition 1 hold. Components $\mathbf{S}_* \mathbf{K} \mathbf{S}'_*$ pass through both maps unchanged and add the same amount to both traces, so they shrink γ towards zero without changing whether it is zero: γ is the gain relative to the whole error, including the part that no reconciliation can remove. The two measures differ in an important way. The gain from any reconciliation comes only from the incoherent component of the errors, so γ is small whenever that component is small, however strongly the joint map leans on other variables' incoherences. A large κ with a small γ says that joint reconciliation would change the forecasts but not improve them by much on average. Because γ averages over all series, it can hide a larger gain for a particular combination of variables. For a combination $\mathbf{a}' \mathbf{y}$ of interest we therefore also compute

$$\gamma_{\mathbf{a}}(\mathbf{W}) = 1 - \frac{\mathbf{a}' \mathbf{M} \mathbf{W} \mathbf{M}' \mathbf{a}}{\mathbf{a}' \mathbf{M}^{\text{sep}} \mathbf{W} \mathbf{M}^{\text{sep}' \mathbf{a}}}, \quad (9)$$

which is also non-negative for every $\mathbf{a}$ (Section 3).

4.3. Estimation and testing

Both measures are computed from an estimate $\widehat{\mathbf{W}}$, and both are biased in finite samples. Sampling noise pushes them above zero even when the population covariance satisfies Proposition 1. The shrinkage estimator pulls the other way, because its diagonal target has no cross-variable blocks (Section 3), so it shrinks genuine departures towards zero. Figure 3 shows both effects. Under a separable covariance the median of $\widehat{\kappa}$ is about 0.12 at the sample sizes of monthly data and falls only slowly with T , while for the strongest node-varying departure in the Brazilian hierarchy $\widehat{\kappa}$ is below its population value at every sample size, and at $T \leq 200$ it is about the same as, or smaller than, under separability. The estimates $\widehat{\kappa}$ and $\widehat{\gamma}$ are therefore useful as descriptions of what joint reconciliation would do with the estimated covariance, but their raw values cannot be read as evidence that the population conditions fail.

A direct test of the conditions follows from the regression reading of Proposition 1. Given its own incoherences $\mathbf{z}_j = \mathbf{C} \mathbf{e}_j$, the errors of variable j are determined by their bottom-level part $\mathbf{e}_j^b$, since the aggregate errors equal $\mathbf{z}_j + \mathbf{A} \mathbf{e}_j^b$. Condition (d) says that, in the multivariate regression

$$\mathbf{e}_{j,t}^b = \boldsymbol{\alpha}_j + \mathbf{B}'_{jj} \mathbf{z}_{j,t} + \sum_{k \neq j} \mathbf{B}'_{jk} \mathbf{z}_{k,t} + \mathbf{u}_{j,t}, \quad (10)$$

the other variables' incoherences add nothing: $\mathbf{B}_{jk} = \mathbf{0}$ for all $k \neq j$. This is a standard linear hypothesis in a multivariate regression with n_b responses and $(m-1)n_a$ restrictions per response. We test it with Wilks' Λ , using Rao's F approximation to its null distribution (Rao, 1951), applied to the one-step residuals of the base models. With m variables there are m such tests, which we combine with a Bonferroni correction. The test uses the sample covariance, so it is unaffected by the shrinkage that biases $\widehat{\kappa}$ and $\widehat{\gamma}$. It needs $T \geq n_b + mn_a + 1$, and like any normal-theory test it assumes serially uncorrelated residual vectors, so it should be applied to residuals of adequate base models.

We also tried calibrating $\widehat{\kappa}$ and $\widehat{\gamma}$ by bootstrap, generating null data from a covariance that satisfies Proposition 1 and is fitted to the data. Such procedures were badly mis-sized when the number of series is large relative to T , because any null covariance built from $\widehat{\mathbf{W}}$ inherits its estimation error, and the regression test avoids simulation altogether. Figure 4 reports its size and power for the covariance

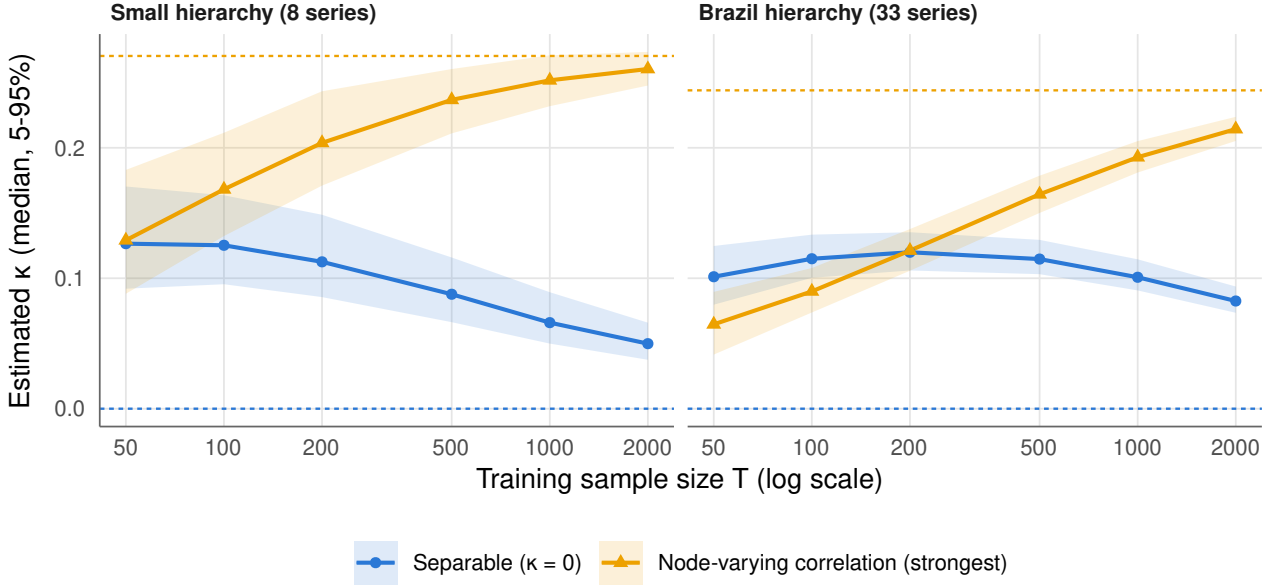


Figure 3: Sampling distribution of $\hat{\kappa}$ (median and 5–95% range over 1000 replications) under a separable covariance and under the strongest node-varying correlation case, with the population values dashed.

families of Section 5. Under both separable and separable-plus-invisible covariances the rejection rate at the 5% level ranges from 4.2% to 5.8% at every sample size, in both hierarchies. Power grows with the size of the departure and with T ; the smallest departures, with population gains below 1%, are detected reliably only with around a thousand observations.

5. Simulations

We use two experiments. The first simulates base forecast errors directly, so that the population error covariance is known and the gain from joint reconciliation can be computed exactly. The second simulates time series, fits base forecasting models, and reconciles their forecasts, as in the original design of this study.

5.1. Experiment 1: controlled error covariances

For each of two variables, the base forecast errors are $\mathbf{e}_j = \mathbf{S}\mathbf{u}_j + \mathbf{v}_j$. The coherent component $\mathbf{S}\mathbf{u}_j$ is invisible to reconciliation (Corollary 1). The incoherent component $\mathbf{v}_j$ has independent entries across series, with variance λ_{ij} for series i and correlation ρ_i between the two variables at series i . With this structure, condition (e) of Proposition 1 holds exactly when $d_i = \rho_i(\lambda_{i2}/\lambda_{i1})^{1/2}$ is the same for every series. We consider five families of covariance matrices, each indexed by a parameter that controls the departure from the conditions of Proposition 1. Let a_i be the number of bottom-level series summed by series i .

F0, separable. $\lambda_{i1} = \lambda_{i2} = a_i$ and $\rho_i = \rho$, with ρ from -0.9 to 0.9 .

F1, separable plus invisible. F0 with $\rho = 0.7$, plus a non-separable coherent component $\mathbf{S}_* \mathbf{K} \mathbf{S}_*'$ of increasing size.

F2, variable-specific profiles. $\lambda_{i1} = a_i$, $\lambda_{i2} = a_i^{1+\delta}$ and $\rho_i = 0.7$. As δ grows, the second variable's incoherent errors become increasingly concentrated at the aggregate levels.

F3, node-varying correlation. $\lambda_{i1} = \lambda_{i2} = a_i$ and $\rho_i = 0.3 + \delta\omega_i$, where $\omega_i \in [-1, 1]$ is a fixed pattern that alternates in sign within each level.

F4, OLS case. $\mathbf{W} = \mathbf{S}_* \mathbf{K} \mathbf{S}_*' + \tau \mathbf{I}$.

F0, F1 and F4 satisfy Proposition 1; F2 and F3 do not when $\delta > 0$. We use two hierarchies: the eight-series hierarchy of Figure 1, and the 33-series hierarchy of the Brazilian application (Section 6).

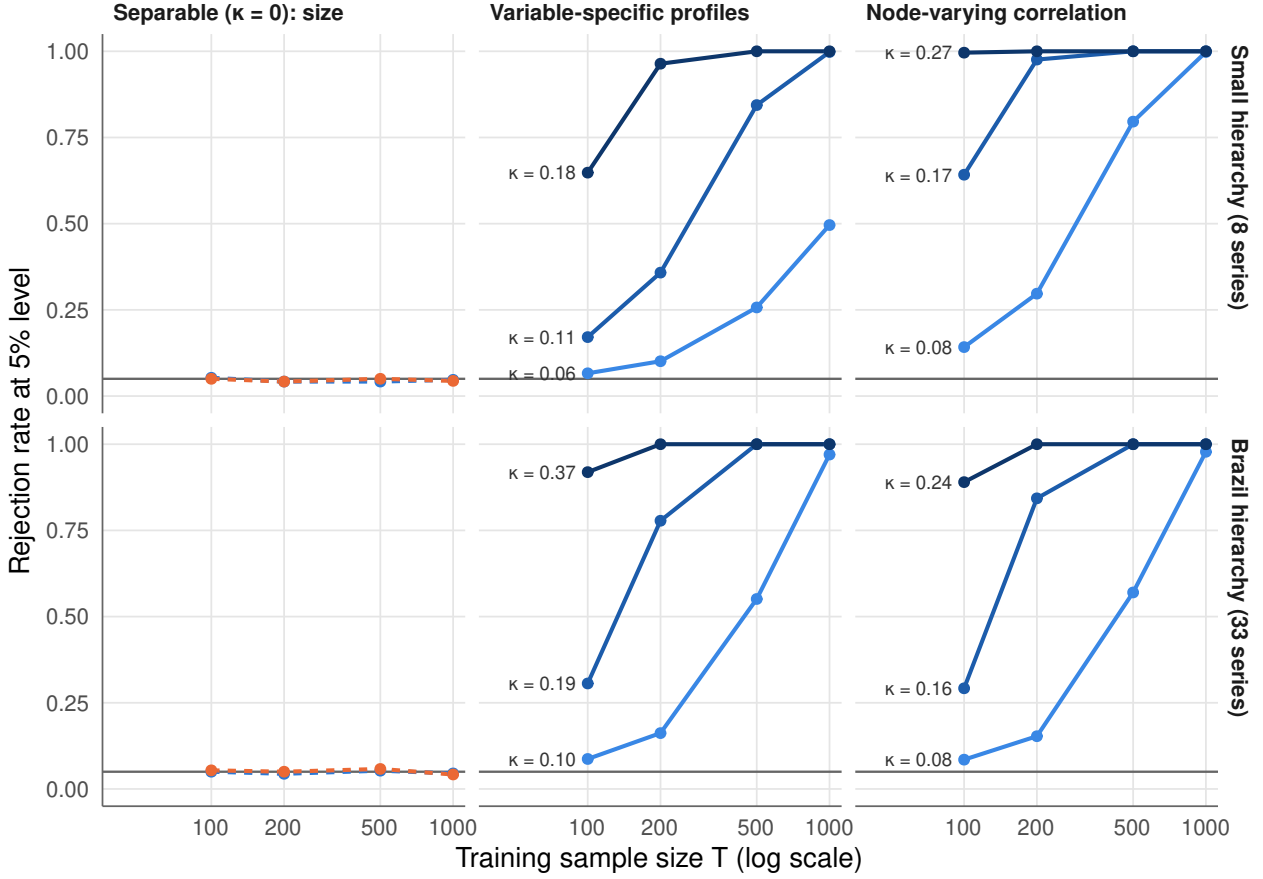


Figure 4: Rejection rate at the 5% level of the test of condition (d), by sample size (1000 replications). For the separable and separable-plus-invisible covariances the rejection rate is the size of the test.

Population results. Figure 5 shows the exact MSE reduction from joint rather than separate reconciliation, $\gamma(\mathbf{W})$, against $\kappa(\mathbf{W})$. For F0, F1 and F4 both are zero to machine precision, whatever the strength of cross-variable correlation, and for F4 joint MinT coincides with OLS reconciliation. For F2 and F3 the gain rises with κ , but it is moderate: at most 15.7% for variable-specific profiles in the Brazilian hierarchy, where κ reaches 0.59, and at most 5.9% for node-varying correlations (in the small hierarchy). Departures from the conditions of Proposition 1 have to be large before joint reconciliation gains more than a few per cent.

Estimated covariances. In practice $\mathbf{W}$ is estimated, and the joint estimate has $m^2 = 4$ times as many entries as each separate one. For each family and hierarchy we drew $T \in \{50, 100, 200, 500, 1000, 2000\}$ error vectors from $N(\mathbf{0}, \mathbf{W})$, estimated the covariance by shrinkage (4), and computed the exact expected loss $\text{tr}(\widehat{\mathbf{M}}\mathbf{W}\widehat{\mathbf{M}}')$ of the resulting maps, over 1000 replications. Figure 6 shows the ratio of expected loss for joint to separate reconciliation. Under separability the joint map is never better, and it is worse at small sample sizes: by 2.9% at $T = 50$ in the small hierarchy and 1.6% in the Brazilian one. That is the price of estimating cross-variable blocks that carry no information. When the population gain is small, the estimated joint map does not recover it until T is in the hundreds or thousands; for the weakest node-varying departure in the Brazilian hierarchy, whose population gain is 0.5%, the estimated joint map gains only 0.1% even at $T = 2000$. Only the largest departures give a clear gain at sample sizes typical of monthly data.

5.2. Experiment 2: fitted base models

The second experiment follows the original design of this study. The two bottom-level series at node i are $\mathbf{x}_{i,t} = \boldsymbol{\mu}_{i,t} + \boldsymbol{\eta}_{i,t}$, with a deterministic quarterly component $\boldsymbol{\mu}_{i,t}$ and a bivariate VAR(1) $\boldsymbol{\eta}_{i,t} = \boldsymbol{\Phi}_i \boldsymbol{\eta}_{i,t-1} + \boldsymbol{\varepsilon}_{i,t}$, aggregated over the eight-series hierarchy of Figure 1. We use six scenarios. In three controls every node has the same dynamics and the innovation covariances have the form

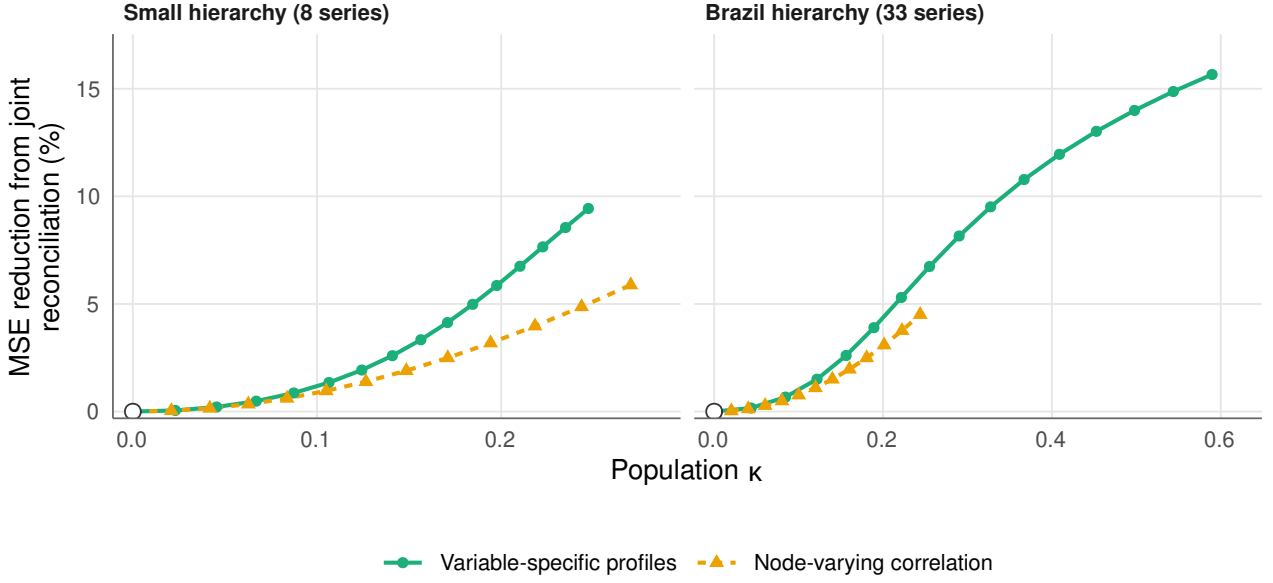


Figure 5: Experiment 1: population MSE reduction from joint rather than separate reconciliation against κ . The separable (F0), separable-plus-invisible (F1) and OLS (F4) families all lie at the origin.

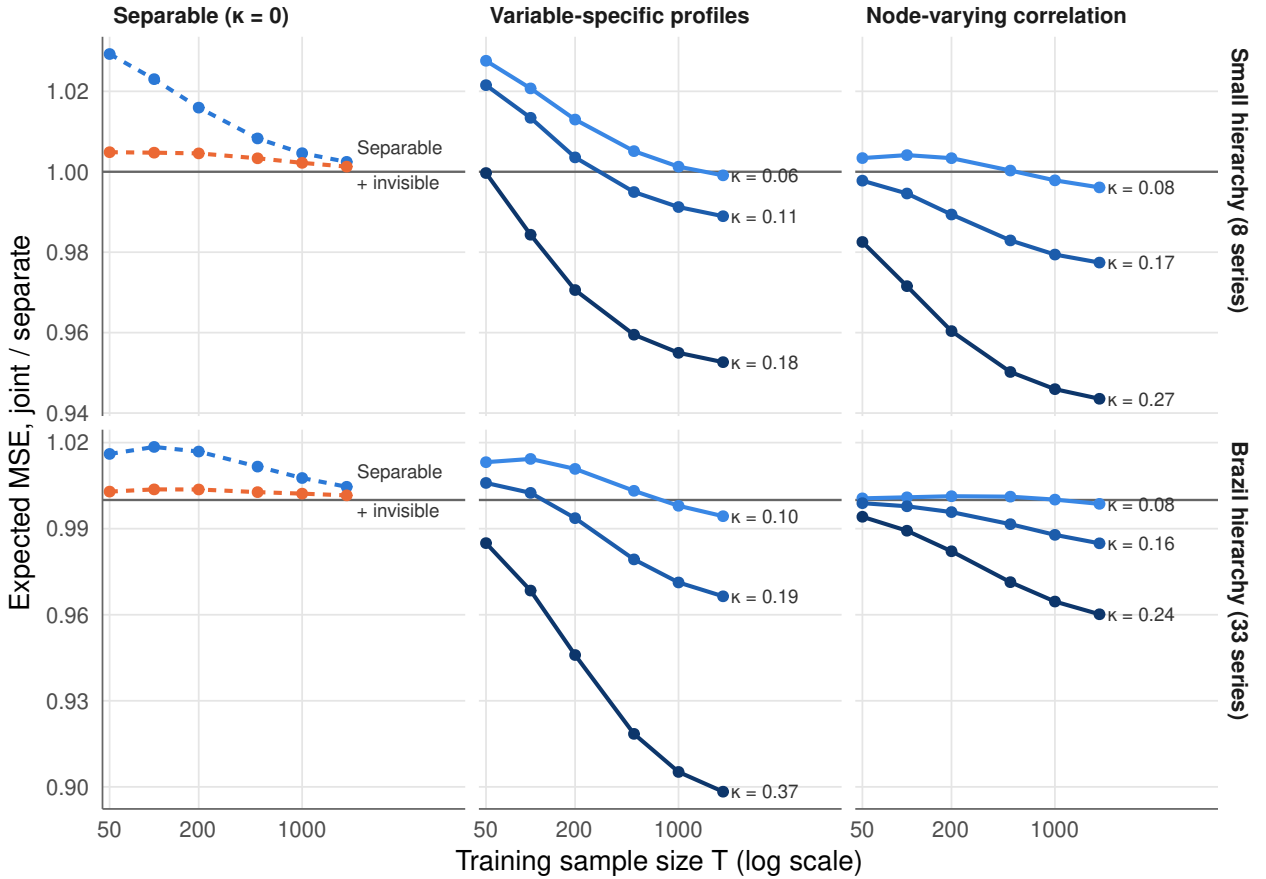


Figure 6: Experiment 1: expected MSE of joint relative to separate reconciliation with shrinkage estimates of $\mathbf{W}$, by training sample size. Values below one favour joint reconciliation. Labels give the population κ .

$\text{Cov}(\varepsilon_{i,t}, \varepsilon_{i',t}) = \sigma_{ii'} \mathbf{R}$ of Proposition 2: cross-variable correlation 0.7 in $\mathbf{R}$, correlation -0.7 , and variable-specific dynamics ($\Phi = \text{diag}(0.9, 0.2)$). By Proposition 2, all three have coherent population forecasts. The other three break node invariance in a way that differs between the variables: a cross-

variable innovation correlation that varies across nodes from -0.6 to 0.9 ; a different node covariance for each variable; and dynamics that vary across nodes in opposite directions for the two variables.

For each scenario we computed the exact one-step error covariance of the population-optimal univariate AR(20) forecasts of all 16 series. Table 1 gives its incoherent share, κ and γ . The controls are coherent to machine precision. In the other scenarios the incoherent component is only 0.4–1.5% of the error variance. So although κ is large (0.65–0.97), the population gain from joint reconciliation is only 0.5–2.4%.

We then simulated 500 series of length $T + 12$ for $T = 108$ and $T = 400$, fitted automatically selected ARIMA models to every series and a bivariate VAR at every node, and reconciled their forecasts for horizons 1–12. Reconciliation improves substantially on the ARIMA base forecasts (by 6–11% in MSE), but joint and separate reconciliation are nearly indistinguishable. With $T = 108$ the ratio of joint to separate MSE lies between 0.996–1.006 for ARIMA base forecasts, and with VAR base forecasts the two never differ by more than 0.5%.

With $T = 400$ and ARIMA base forecasts, joint reconciliation is no worse in any scenario and is better by up to 2.1% in the controls and up to 1.5% in the non-separable scenarios, so the gain is at least as large in the controls. This is not a contradiction of Proposition 2. The fitted models differ from node to node and are estimated from correlated data, so their forecasts are incoherent even when the population forecasts are not, and the cross-variable structure of this estimation error need not satisfy Proposition 1. Using the diagonal blocks of the joint covariance estimate for separate reconciliation changes the MSE by at most 0.2%, so the gain comes from the cross-variable blocks rather than from differences in shrinkage. In these designs, what joint reconciliation exploits is the structure of the base models’ estimation error, not the structure of the data-generating process, and the resulting gains are small.

Table 1: Experiment 2. Population quantities for the optimal AR forecasts: incoherent share of error variance (%), κ , and MSE reduction from joint reconciliation (%). Fitted base models: MSE of joint relative to separate reconciliation, averaged over 500 replications, 16 series and horizons 1–12, with the standard error of the log ratio in parentheses; mean plug-in gain $\hat{\gamma}$ (% , ARIMA).

Scenario	T	Population			Joint / separate			
		Incoh.	κ	Gain	ARIMA	VAR	$\hat{\gamma}$	
Separable, $\rho = 0.7$	108	0.00	0.00	0.00	0.997 (0.003)	1.002 (0.001)	1.7	
Separable, $\rho = 0.7$	400	0.00	0.00	0.00	0.979 (0.004)	0.999 (0.001)	1.8	
Separable, $\rho = -0.7$	108	0.00	0.00	0.00	0.996 (0.003)	1.004 (0.001)	2.1	
Separable, $\rho = -0.7$	400	0.00	0.00	0.00	0.987 (0.004)	0.999 (0.001)	2.3	
Variable-specific dynamics	108	0.00	0.00	0.00	1.006 (0.002)	1.000 (0.001)	1.8	
Variable-specific dynamics	400	0.00	0.00	0.00	0.996 (0.003)	0.997 (0.001)	2.0	
Node-varying correlation	108	0.42	0.79	2.14	0.998 (0.003)	1.005 (0.001)	1.5	
Node-varying correlation	400	0.42	0.79	2.14	0.991 (0.003)	0.998 (0.001)	1.7	
Variable-specific node covariance	108	1.28	0.97	2.37	0.998 (0.003)	1.003 (0.002)	1.6	
Variable-specific node covariance	400	1.28	0.97	2.37	0.990 (0.003)	1.001 (0.001)	1.6	
Node-varying dynamics	108	1.52	0.65	0.46	0.999 (0.003)	1.000 (0.001)	1.9	
Node-varying dynamics	400	1.52	0.65	0.46	0.985 (0.003)	0.995 (0.001)	1.9	

6. Application: admissions and dismissals in Brazil

6.1. Data

Brazil’s General Register of Employed and Unemployed (CAGED) records every admission and dismissal of formally employed workers. The two variables share a geographical hierarchy: 26 states and the Federal District (27 federative units) aggregate to five regions, which aggregate to the country (Figure 7). With two variables this gives $m = 2$, $n = 33$ and $n_b = 27$, so 66 series in all.

We built the monthly series directly from the CAGED microdata published by the Labour Statistics Dissemination Program (PDET) of the Ministry of Labour and Employment, counting admissions and dismissals by federative unit and month. Records without an identified federative unit (fewer than



Figure 7: Brazilian regions and federative units.

0.1% before 2020) are excluded. Microdata are available from January 2007. Twenty-two of the 156 monthly archives for 2007–2019 are damaged on the PDET server; for those months we used an earlier extract of the same data, which agrees with the microdata to within one record in every federative unit for all intact months. National totals agree with the official series published by IpeaData. The code that downloads the microdata and builds the series is part of the replication material.

In January 2020 CAGED was replaced by the Novo CAGED, which draws on different administrative sources, and the COVID-19 pandemic began soon after. Both events change the joint behaviour of admissions and dismissals, and would contaminate an estimate of their forecast error covariance. Our main analysis therefore uses January 2007 to December 2019 (156 months).

Figure 8 shows the national series. Admissions and dismissals move together, with strong seasonality: December brings large net job losses every year. Net employment change was positive until 2014, negative through the recession of 2015–2016, and positive again from 2017.

6.2. Forecasting set-up

We forecast all 66 series from rolling origins with an expanding window, starting with 108 months of training data (January 2007 to December 2015) and moving the origin forward one month at a time. This gives 48 origins, and 37 of them have all twelve horizons within the sample. Base forecasts come from an ARIMA model selected automatically for each series (Hyndman and Athanasopoulos, 2021) and from a bivariate VAR for admissions and dismissals at each node. We compare the base forecasts with OLS, structurally weighted (WLS) and MinT reconciliation. MinT is applied separately to each variable, jointly to both, and separately using the diagonal blocks of the joint covariance estimate, which isolates the effect of the cross-variable blocks from that of the different shrinkage intensities.

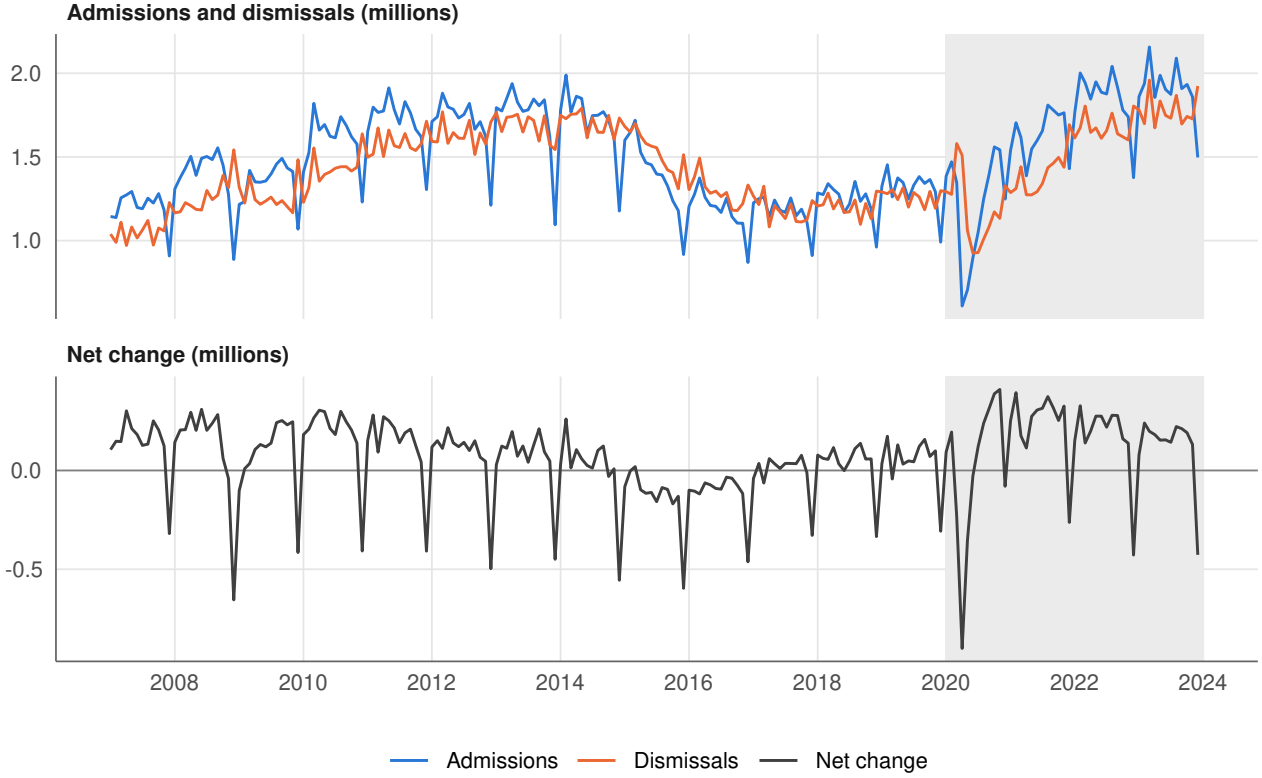


Figure 8: Monthly admissions, dismissals and net employment change in Brazil. The shaded period (Novo CAGED and the COVID-19 pandemic) is not used in the main analysis.

6.3. Diagnostics

Table 2 reports the diagnostics computed from the one-step residuals of models fitted to the whole of 2007–2019. For the ARIMA base forecasts the estimated error covariance is close to separable by the generic measure (nearest-Kronecker error 0.07), and joint reconciliation would place moderate weight on the other variable’s incoherences ($\hat{\kappa} = 0.34$) for a plug-in gain of only 2.0% in mean squared error. The likelihood ratio test rejects the conditions of Proposition 1 at the 5% level ($p = 0.020$), so the departure is not pure estimation noise, but it is small. The incoherent component is 17% of the one-step error variance. The VAR residuals give larger $\hat{\kappa}$ and a much smaller p -value, but they are strongly autocorrelated for almost every series, which invalidates the test; the per-node VARs, fitted without seasonal terms, probably do not capture the monthly seasonality.

The plug-in gain for net employment change is larger than for the individual series: 4.9% for the national net change, 4.5% for the regions and 3.2% for the states, on average.

Table 2: Application: diagnostics from the one-step residuals over 2007–2019. Estimated $\hat{\kappa}$ and plug-in gain $\hat{\gamma}$ (%); Rao F statistics for the other variable’s incoherences in the regression of each variable’s errors (162 and 696 degrees of freedom for ARIMA, 162 and 666 for VAR), and the Bonferroni p -value; incoherent share of error variance (%); relative error of the nearest Kronecker product.

Base model	T	$\hat{\kappa}$	$\hat{\gamma}$	F		p	Incoh.	Kron.
				Adm.	Dis.			
ARIMA	156	0.338	2.05	1.29	1.32	0.020	17.1	0.068
VAR	151	0.602	2.30	2.44	2.95	<0.001	13.0	0.082

6.4. Forecast accuracy

Table 3 gives the mean squared error of each method relative to the ARIMA base forecasts. Separate MinT reconciliation improves on the base forecasts at every level and for both variables, and it is clearly better than OLS and structural WLS reconciliation, which damage the state-level forecasts. Joint and

separate MinT are close: the ratio of their mean squared errors ranges from 0.986–1.019 across the six combinations of variable and level. Joint reconciliation is slightly better for dismissals at the national and regional levels and slightly worse elsewhere. The averages hide a horizon pattern (Supplementary Table S3): for national dismissals, joint reconciliation is worse at short horizons and better at long ones. There is no systematic gain from reconciling the two variables together, consistent with the small departure found by the diagnostics. With VAR base forecasts (Table 4), joint reconciliation is better than separate reconciliation at every level (ratios 0.951–0.996), but the VAR base forecasts are poorer to begin with: their mean squared error is 50% higher than that of the ARIMA base forecasts, on average over the series. This is consistent with the lesson of Experiment 2 that the cross-variable structure joint reconciliation exploits can come from the errors of the base models themselves. With ETS base forecasts (Supplementary Table S2) the ratios of joint to separate MSE range from 0.951–1.016: joint reconciliation is better for admissions at every level and roughly neutral for dismissals.

Table 3: Application: MSE relative to base forecasts (ARIMA), geometric mean over the series at each level, averaged over 48 forecast origins (37 with all twelve horizons) and horizons 1–12.

Method	Admissions			Dismissals		
	Total	Regions	States	Total	Regions	States
Base	1.000	1.000	1.000	1.000	1.000	1.000
OLS	0.955	1.439	1.804	0.977	1.151	1.491
WLS (structural)	0.894	1.105	1.308	0.945	0.960	1.170
MinT, separate	0.878	0.930	0.984	0.844	0.880	0.940
MinT, separate (joint estimate)	0.873	0.928	0.981	0.853	0.886	0.943
MinT, joint	0.880	0.937	1.003	0.832	0.872	0.952

Table 4: Application: MSE relative to base forecasts (VAR), geometric mean over the series at each level, averaged over 48 forecast origins (37 with all twelve horizons) and horizons 1–12.

Method	Admissions			Dismissals		
	Total	Regions	States	Total	Regions	States
Base	1.000	1.000	1.000	1.000	1.000	1.000
OLS	1.012	0.954	1.335	0.996	1.018	1.184
WLS (structural)	1.033	0.923	1.139	0.981	0.969	1.075
MinT, separate	1.108	0.985	1.043	0.902	0.909	0.988
MinT, separate (joint estimate)	1.068	0.954	1.020	0.912	0.913	0.989
MinT, joint	1.076	0.937	1.002	0.898	0.902	0.982

6.5. Net employment change

Net employment change is a linear combination of the two variables, so the argument of Section 3 suggests that joint reconciliation may help it even when it does not help either variable alone. Table 5 confirms this. For point forecasts, joint reconciliation reduces the mean squared error of net change relative to separate reconciliation by 18% at the national level and 8% for the regions, though only 1% for the states. Joint reconciliation makes the errors in admissions and dismissals more positively correlated (a correlation of 0.72 against 0.63 for separate reconciliation, pooled over all series and horizons), and they cancel in the difference. Part of the gain comes from the different shrinkage intensity of the joint covariance estimate, as the row for separate reconciliation with the joint estimate shows. At the national and regional levels most of it comes from the cross-variable blocks; the small gain for the states comes entirely from the shrinkage.

For probabilistic forecasts we generated 1000 sample paths of all 66 series at each origin from the fitted ARIMA models, driven by resampled one-step residuals, and reconciled every path. Resampling each variable’s innovations independently discards the dependence between admissions and dismissals; resampling whole residual vectors keeps it. Keeping the dependence reduces the CRPS of net change by 15% nationally, 10% for the regions and 5% for the states, using separate reconciliation of each variable. Reconciling the joint paths jointly adds a further gain at the aggregate levels, to 21% and 14% respectively. The largest single effect comes from modelling the dependence in the base forecasts; joint reconciliation adds to it for the aggregate net change.

Table 5: Application: forecasts of net employment change (admissions minus dismissals), ARIMA base models, geometric mean over the series at each level, 48 origins (37 with all twelve horizons), horizons 1–12. Top: MSE of point forecasts relative to separate MinT. Bottom: CRPS from 1000 sample paths relative to independent innovations with separate reconciliation.

	Total	Regions	States
<i>Point forecasts (MSE)</i>			
MinT, separate	1.000	1.000	1.000
MinT, separate (joint estimate)	0.949	0.973	0.991
MinT, joint	0.818	0.919	0.991
<i>Sample paths (CRPS)</i>			
Independent innovations, separate reconciliation	1.000	1.000	1.000
Joint innovations, separate reconciliation	0.853	0.901	0.950
Joint innovations, joint reconciliation	0.787	0.863	0.945

7. Discussion

When several variables are observed over the same hierarchy and no constraint links them, joint MinT reconciliation can improve on reconciling each variable separately only through the covariance between one variable’s reconciled errors and another variable’s incoherences. Cross-variable correlation in the base forecast errors is not enough. Correlation that is common to the whole hierarchy (the separable case) and correlation in the coherent component of the errors are both invisible to reconciliation. What remains is the part of the dependence that differs across the hierarchy in a way that differs between variables, and that part is typically a small fraction of the forecast error variance.

This has three practical consequences.

First, simulation designs need care. A design in which every node shares the same dynamics produces coherent population forecasts (Proposition 2), so any difference between joint and separate reconciliation comes from estimation error in the base models. Experiment 2 shows that this estimation error can itself have exploitable cross-variable structure, but that the resulting gains are small and appear only with long series. Varying the cross-variable correlation in such a design, however strongly, says nothing about the value of joint reconciliation for the data-generating process.

Second, check before reconciling jointly. The diagnostic of Section 4 estimates how much joint reconciliation would change the forecasts (κ) and how much it would improve them (γ), and a likelihood ratio test of condition (d) says whether the departure is distinguishable from estimation noise. When they are not, separate reconciliation is preferable: it needs only the diagonal blocks of the error covariance, which are estimated more precisely, and Experiment 1 shows that estimating the cross-variable blocks costs accuracy when they carry no information.

Third, think about the quantity being forecast. Many quantities of interest are functions of several variables, such as net employment change. Their forecast distributions depend on the joint distribution of the base forecasts, and the application shows that ignoring that dependence is costly. Because reconciliation is linear, the reconciled joint distribution is the base joint distribution pushed through the reconciliation map, so separate reconciliation of jointly simulated base paths already captures the cross-variable dependence, and it coincides with joint reconciliation whenever the conditions of Proposition 1 hold. When they fail, joint reconciliation can still help such combinations substantially while barely changing each variable’s average accuracy, as it did for the national and regional net change in Brazil. The plug-in gain $\gamma_{\mathbf{a}}$ for the combination of interest is a better guide than the average gain γ .

Joint reconciliation is genuinely needed when a constraint does link the variables. If net employment change were forecast as a series in its own right, the constraint that it equals admissions minus dismissals would couple the two variables, and no variable-by-variable reconciliation could satisfy it. Such signed linear constraints fit the general framework of Girolimetto and Di Fonzo (2024). With them $\mathbf{C}_*$ is no longer block diagonal, separate reconciliation is not an option, and the question addressed here does not arise.

Several extensions are natural. The shrinkage estimator treats cross-variable covariances like any other off-diagonal entry; an estimator that shrinks towards the null set of Proposition 1 would interpolate between separate and joint reconciliation using the data to choose how far to move. The diagnostic could also be applied pairwise to decide which groups of variables to reconcile together when there are many. Finally, the same questions arise for temporal and cross-temporal hierarchies with several variables, where the characterisation of Section 3 extends directly.

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Appendix A. Proofs

Proof of Proposition 1. Let $\mathbf{Q} = \mathbf{C}_* \mathbf{W} \mathbf{C}'_*$, with blocks $\mathbf{Q}_{jk} = \mathbf{C} \mathbf{W}_{jk} \mathbf{C}'$. Because $\mathbf{W}$ is positive definite and $\mathbf{C}_*$ has full row rank, $\mathbf{Q}$ is invertible, so $\mathbf{G}$ is the unique solution of $\mathbf{G} \mathbf{Q} = \mathbf{W} \mathbf{C}'_*$. Block (j, k) of the right-hand side is $\mathbf{W}_{jk} \mathbf{C}'$.

(c) $\Rightarrow$ (a). Let $\mathbf{G} = \text{diag}(\mathbf{H}_1, \dots, \mathbf{H}_m)$. Block (j, j) of $\mathbf{G} \mathbf{Q} = \mathbf{W} \mathbf{C}'_*$ gives $\mathbf{H}_j \mathbf{C} \mathbf{W}_{jj} \mathbf{C}' = \mathbf{W}_{jj} \mathbf{C}'$, so $\mathbf{H}_j = \mathbf{G}_j$. Hence $\mathbf{M} = \mathbf{I} - \mathbf{G} \mathbf{C}_* = \text{diag}(\mathbf{I} - \mathbf{G}_j \mathbf{C}) = \mathbf{M}^{\text{sep}}$.

(a) $\Rightarrow$ (b) is immediate.

(b) $\Rightarrow$ (c). $\mathbf{G} \mathbf{C}_* = \mathbf{I} - \mathbf{M}$ is block diagonal. Since $\mathbf{C}_*$ is block diagonal with full row rank, it has the block diagonal right inverse $\mathbf{C}'_*(\mathbf{C}_* \mathbf{C}'_*)^{-1}$, and $\mathbf{G} = (\mathbf{G} \mathbf{C}_*) \mathbf{C}'_*(\mathbf{C}_* \mathbf{C}'_*)^{-1}$ is block diagonal.

(c) $\Leftrightarrow$ (d). By the argument for (c) $\Rightarrow$ (a), $\mathbf{G}$ is block diagonal if and only if $\mathbf{G} = \mathbf{G}^{\text{sep}} = \text{diag}(\mathbf{G}_1, \dots, \mathbf{G}_m)$. By uniqueness, this holds if and only if $\mathbf{G}^{\text{sep}} \mathbf{Q} = \mathbf{W} \mathbf{C}'_*$. Blocks (j, j) of this equation hold by the definition of $\mathbf{G}_j$. Blocks (j, k) with $j \neq k$ read $\mathbf{G}_j \mathbf{C} \mathbf{W}_{jk} \mathbf{C}' = \mathbf{W}_{jk} \mathbf{C}'$, that is, $(\mathbf{I} - \mathbf{G}_j \mathbf{C}) \mathbf{W}_{jk} \mathbf{C}' = \mathbf{M}_j \mathbf{W}_{jk} \mathbf{C}' = \mathbf{0}$. Finally, $\text{Cov}(\mathbf{M}_j \mathbf{e}_j, \mathbf{C} \mathbf{e}_k) = \mathbf{M}_j \mathbf{W}_{jk} \mathbf{C}'$.

(d) $\Leftrightarrow$ (e). Condition (d) states $\mathbf{W}_{jk} \mathbf{C}' = \mathbf{W}_{jj} \mathbf{C}' \mathbf{B}_{jk}$ with $\mathbf{B}_{jk} = (\mathbf{C} \mathbf{W}_{jj} \mathbf{C}')^{-1} \mathbf{C} \mathbf{W}_{jk} \mathbf{C}'$, so (d) implies (e). Conversely, if $\mathbf{W}_{jk} \mathbf{C}' = \mathbf{W}_{jj} \mathbf{C}' \mathbf{B}$ for some $\mathbf{B}$, then $\mathbf{M}_j \mathbf{W}_{jk} \mathbf{C}' = \mathbf{M}_j \mathbf{W}_{jj} \mathbf{C}' \mathbf{B} = \mathbf{0}$, because $\mathbf{M}_j \mathbf{W}_{jj} \mathbf{C}' = \mathbf{W}_{jj} \mathbf{C}' - \mathbf{G}_j \mathbf{C} \mathbf{W}_{jj} \mathbf{C}' = \mathbf{0}$. $\square$

Proof of Corollary 1. Since $\mathbf{C} \mathbf{S} = \mathbf{0}$, we have $\mathbf{C}_* \mathbf{S}_* = \mathbf{0}$, so $(\mathbf{W} + \mathbf{S}_* \mathbf{K} \mathbf{S}'_*) \mathbf{C}'_* = \mathbf{W} \mathbf{C}'_*$ and $\mathbf{C}_* (\mathbf{W} + \mathbf{S}_* \mathbf{K} \mathbf{S}'_*) \mathbf{C}'_* = \mathbf{C}_* \mathbf{W} \mathbf{C}'_*$. Hence $\mathbf{G}$, and so $\mathbf{M}$, is unchanged. The diagonal blocks change by $\mathbf{S} \mathbf{K}_{jj} \mathbf{S}'$, which leaves each $\mathbf{G}_j$, and so $\mathbf{M}^{\text{sep}}$, unchanged by the same argument. $\square$

Proof of Corollary 2. Here $\mathbf{W}_{jk} = v_{jk} \mathbf{\Sigma}_0$, so $\mathbf{G}_j = \mathbf{\Sigma}_0 \mathbf{C}' (\mathbf{C} \mathbf{\Sigma}_0 \mathbf{C}')^{-1}$ and $\mathbf{M}_j = \mathbf{M}_0$ for every j . Then $\mathbf{W}_{jk} \mathbf{C}' = (v_{jk}/v_{jj}) \mathbf{W}_{jj} \mathbf{C}'$, and condition (e) holds. $\square$

Proof of Proposition 2. Let $\mathbf{s}'_a$ be row a of $\mathbf{S}$, and let $\mathbf{\Gamma}_h$ be the lag- h autocovariance matrix of a VAR(1) with coefficient $\mathbf{\Phi}$ and innovation covariance $\mathbf{R}$. Since $\text{Cov}(\boldsymbol{\eta}_{i,t+h}, \boldsymbol{\eta}_{i',t}) = \sigma_{ii'} \mathbf{\Gamma}_h$, the stochastic component of series (j, a) , $\zeta_{j,a,t} = \sum_i s_{ai} \eta_{i,j,t}$, has autocovariances $\text{Cov}(\zeta_{j,a,t+h}, \zeta_{j,a,t}) = (\mathbf{s}'_a \mathbf{\Sigma} \mathbf{s}_a) [\mathbf{\Gamma}_h]_{jj}$. These are proportional across a , so the best linear predictor of $\zeta_{j,a,t+h}$ from $\zeta_{j,a,t}, \zeta_{j,a,t-1}, \dots$ applies the same linear filter $f_{j,h}$ at every node. A linear filter commutes with aggregation: $f_{j,h}(\zeta_{j,a,\cdot}) = \sum_i s_{ai} f_{j,h}(\eta_{i,j,\cdot})$. The deterministic components also aggregate, so the base forecasts of variable j equal $\mathbf{S}$ times its bottom-level base forecasts and are coherent. The errors of coherent forecasts lie in the column space of $\mathbf{S}_*$, and any $\mathbf{S}_* \mathbf{P}$ with $\mathbf{P} \mathbf{S}_* = \mathbf{I}$ maps a vector $\mathbf{S}_* \mathbf{b}$ to itself. $\square$